

CNuvS Ex. 11 - Jan Lindauer, Paul Feidieker

Task 4

- (a) The first distinct operational risk is the missing redundancy. If for some reason the supercomputer in Frankfurt goes offline (power outage, crashes, cyberattack etc.), no one can access websites using standard URLs anymore, since there's no other way to translate domain names.

The second operational risk is latency and traffic. Users sitting far away from Frankfurt would suffer from massive delays just because of the distance to the server. Another problem would be the combined traffic of the whole world, choking Frankfurts network.

The solution to the redundancy issue is the hierarchical, decentralized structure of DNS that is backed by Anycast routing. The 13 global root servers are duplicated into thousands of instances. If one instance goes out, traffic is just rerouted to the next closest one.

The solution to the latency and traffic issue is the caching logic of DNS. Previously visited IPs are saved locally for some time (TTL) and only a small amount of queries creates global traffic.

[1, p. 21, 24]

Sources

[1] Prof. Dr. Marco Zimmerling, “CNUvS 07: Application Layer.” 2026.